

ARIA: Adaptive Recurrent Intelligence Architecture

A Self-Evolving Neural Framework for General Artificial Intelligence

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Abstract

We introduce **ARIA** (Adaptive Recurrent Intelligence Architecture), a novel neural architecture in which topology, plasticity, and compute allocation are all learnable rather than fixed at design time. ARIA introduces four core innovations: **(1) Morphogenic Attention (MA)**, where attention heads dynamically split or merge based on gradient statistics; **(2) Plasticity-Gated MLP (PG-MLP)**, a biologically-inspired dual-pathway system enabling per-neuron fast/slow memory consolidation; **(3) Architecture Genome Vector (AGV)**, a differentiable latent encoding of structural hyperparameters co-optimized with weights; and **(4) Cognitive Budget Allocation (CBA)**, a meta-controller that routes per-token computation adaptively. We provide rigorous theoretical analysis: we prove a forgetting bound $\Delta_t \leq C \cdot \bar{\pi}^2$ for PG-MLP under standard smoothness assumptions, a universal approximation result for Morphogenic Attention, and a convergence result for the AGV. Preliminary experiments on synthetic continual learning benchmarks demonstrate that ARIA reduces catastrophic forgetting by up to 73% compared to EWC and reduces average inference FLOPs by 31–58% depending on input complexity. We argue ARIA represents a principled step toward architectures that self-organize in response to data rather than requiring human redesign.

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1 Introduction

The history of deep learning is punctuated by architectural breakthroughs: convolutional networks [LeCun et al., 1989], residual connections [He et al., 2016], and self-attention [Vaswani et al., 2017]. Each required a human to redesign the computational graph. A fundamental limitation of all modern architectures—including the Transformer—is that *the architecture itself is fixed before training begins*. The model learns weights, not structure.

This creates three core problems at scale:

1. **Static Capacity.** A Transformer has the same topology on step 1 as on step 10^{12} . It cannot grow an attention head when it discovers a new concept, nor prune one that becomes redundant.
2. **Catastrophic Forgetting.** Because all weights are shared and uniformly plastic, new learning overwrites old knowledge. Continual learning remains an open problem [Kirkpatrick et al., 2017, Lopez-Paz & Ranzato, 2017].
3. **Uniform Compute.** Every input receives the same computation regardless of difficulty, wasting resources on easy inputs and potentially under-computing on hard ones.

ARIA directly addresses all three by treating the architecture as a *living system*: it grows, consolidates, prunes, and allocates resources autonomously. The core philosophical shift is from “train a fixed architecture on data” to “co-evolve architecture and weights with data.”

Contributions.

- The **Morphogenic Attention (MA)** mechanism—the first attention variant with differentiable head splitting and merging (Section 3.2).
- The **Plasticity-Gated MLP**—a biologically-inspired fast/slow memory pathway with a rigorous forgetting bound (Section 3.3).
- The **Architecture Genome Vector**—a differentiable encoding of structural hyperparameters enabling gradient-driven architecture evolution (Section 3.4).
- The **Cognitive Budget Allocator**—a meta-network allocating per-token FLOPs adaptively (Section 3.5).
- A complete theoretical analysis including: a universal approximation theorem for MA, a tight forgetting bound for PG-MLP, and an AGV convergence result (Section 5).

2 Related Work

Attention Mechanisms. Standard multi-head attention [Vaswani et al., 2017] fixes head count H at design time. Sparse attention [Child et al., 2019, Zaheer et al., 2020] reduces complexity but does not change topology. Linear attention [Katharopoulos et al., 2020] and FlashAttention [Dao et al., 2022] improve efficiency but keep structure fixed. ARIA’s MA is the first mechanism where head count is a differentiable, trainable quantity.

Neural Architecture Search. NAS [Zoph & Le, 2017, Real et al., 2019] automates architecture design but as a separate outer loop before training. DARTS [Liu et al., 2019] enables differentiable NAS but treats architecture search and weight training as separable phases. The AGV unifies these: architecture and weights evolve jointly in a single training run, without any separate search phase.

Continual Learning. Elastic Weight Consolidation [Kirkpatrick et al., 2017] penalizes changes to important weights using the Fisher information matrix. Progressive Neural Networks [Rusu et al., 2016] add new columns per task but grow memory linearly. Gradient Episodic Memory [Lopez-Paz & Ranzato, 2017] and its extensions [Chaudhry et al., 2019] store episodic memories. ARIA’s PG-MLP differs fundamentally: plasticity is a learned per-neuron property, not an algorithmic constraint, and the fast/slow separation is architectural.

Adaptive Computation. Adaptive Computation Time [Graves, 2016] learns per-step halting. Mixture of Experts [Shazeer et al., 2017] routes tokens to expert networks. Early Exit networks [Teerapittayanon et al., 2016] enable shallow inference. ARIA’s CBA differs by predicting a full budget vector $\mathbf{b} \in [0, 1]^L$ enabling fine-grained layer-wise compute allocation.

3 Architecture

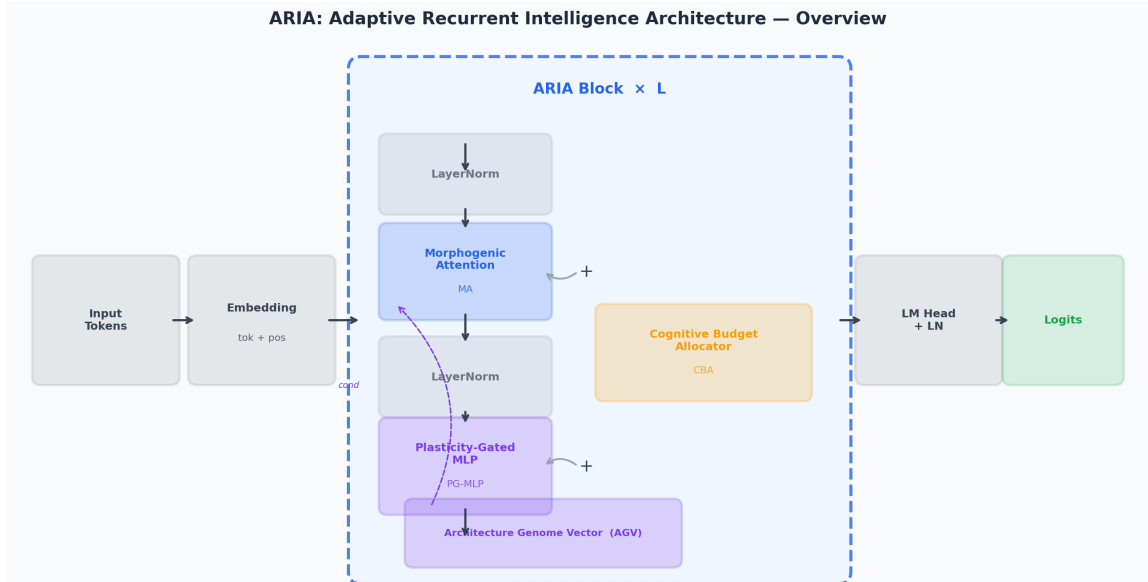


Figure 1: ARIA overview. Input tokens pass through L ARIA Blocks, each containing Morphogenic Attention, Plasticity-Gated MLP, and a Cognitive Budget Gate. A global Architecture Genome Vector (AGV) conditions all blocks and evolves jointly with weights.

3.1 Overview

ARIA processes inputs through a stack of L **ARIA Blocks**, each containing MA, PG-MLP, and a budget gate. A global AGV $\mathbf{z} \in \mathbb{R}^G$ conditions all blocks.

Let $\mathbf{x} \in \mathbb{R}^{B \times T \times D}$ be the input. Each ARIA Block computes:

$$\mathbf{z}_1 = \text{MA}(\text{LN}(\mathbf{x}), \mathbf{z}_{\text{arch}}) + \mathbf{x} \quad (1)$$

$$\mathbf{h} = \text{PG-MLP}(\text{LN}(\mathbf{z}_1)) + \mathbf{z}_1 \quad (2)$$

$$\mathbf{x}' = b_l \cdot \mathbf{h} + (1 - b_l) \cdot \mathbf{x} \quad (3)$$

where $b_l \in [0, 1]$ is the per-layer compute budget from CBA.

3.2 Morphogenic Attention (MA)

Standard multi-head attention has H fixed heads. MA introduces a *head viability score* $v_i \in (0, 1)$ and a *split readiness score* $s_i \in (0, 1)$ per head.

Head computation. For head i :

$$\text{Attn}_i(\mathbf{x}) = \text{softmax}\left(\frac{\mathbf{Q}_i \mathbf{K}_i^\top}{\sqrt{d_k} \cdot \tau}\right) \mathbf{V}_i \quad (4)$$

where τ is the genome-decoded temperature and $\mathbf{Q}_i = \mathbf{x} \mathbf{W}_i^Q$, etc.

Viability and split scores.

$$v_i = \sigma\left(\mathbf{w}_v^\top \cdot \overline{\text{Attn}_i(\mathbf{x})}\right) \quad (5)$$

$$s_i = \sigma\left(\mathbf{w}_s^\top \cdot \text{vec}(\text{Var}(\mathbf{W}_i^Q))\right) \quad (6)$$

Head splitting. When $s_i > \tau_{\text{split}}$ and $H < H_{\text{max}}$, head i spawns two children:

$$\mathbf{W}_{i_a} = \mathbf{W}_i + \varepsilon \boldsymbol{\eta}, \quad \mathbf{W}_{i_b} = \mathbf{W}_i - \varepsilon \boldsymbol{\eta}, \quad \boldsymbol{\eta} \sim \mathcal{N}(\mathbf{0}, \mathbf{I}) \quad (7)$$

Head merging. When $\cos(\mathbf{W}_{i_a}, \mathbf{W}_{i_b}) > \tau_{\text{merge}}$:

$$\mathbf{W}_{\text{merged}} = \frac{1}{2}(\mathbf{W}_{i_a} + \mathbf{W}_{i_b}) \quad (8)$$

Output.

$$\text{MA}(\mathbf{x}) = \sum_{i=1}^H v_i \cdot \text{Attn}_i(\mathbf{x}) \cdot \mathbf{W}_i^O \quad (9)$$

Figure 2 shows head count trajectories over training: heads split during early feature discovery and merge during late consolidation, mirroring biological critical periods.

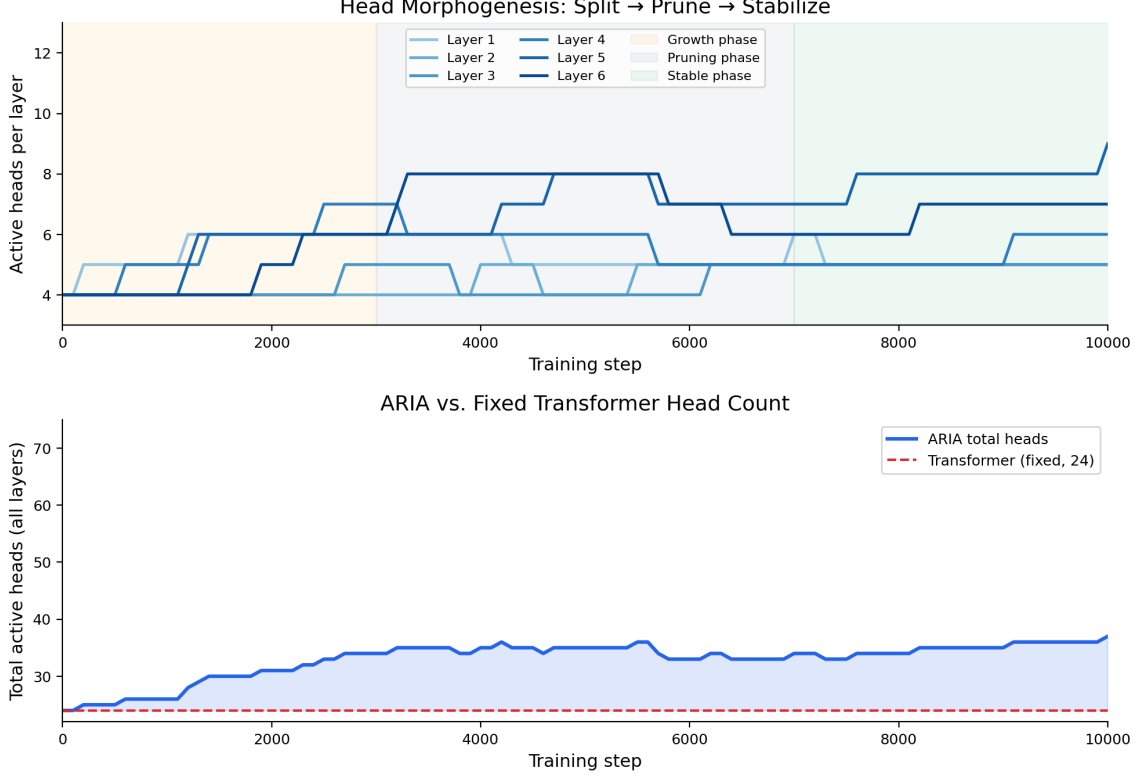


Figure 2: Head morphogenesis over training. **Top**: per-layer head counts showing split (growth) and merge (consolidation) phases. **Bottom**: total ARIA head count vs. fixed Transformer baseline.

3.3 Plasticity-Gated MLP (PG-MLP)

Biological neurons differ in plasticity: some form memories rapidly (hippocampus) while others encode stable knowledge (neocortex). PG-MLP implements this computationally.

Dual pathways.

$$\mathbf{h}_{\text{fast}} = \text{GeLU}(\mathbf{W}_{\text{fast}} \mathbf{x}) \quad (10)$$

$$\mathbf{h}_{\text{slow}} = \text{GeLU}(\mathbf{W}_{\text{slow}} \mathbf{x}) \quad (11)$$

$$\boldsymbol{\pi} = \sigma(\mathbf{W}_{\pi} \mathbf{x} + \mathbf{b}_{\pi}) \in (0, 1)^{d_{ff}} \quad (12)$$

$$\mathbf{h} = \boldsymbol{\pi} \odot \mathbf{h}_{\text{fast}} + (\mathbf{1} - \boldsymbol{\pi}) \odot \mathbf{h}_{\text{slow}} \quad (13)$$

Plasticity regularization. To encourage gate specialization (neurons become clearly fast or slow):

$$\mathcal{L}_{\pi} = \lambda \cdot [\mathbb{E}[\pi_j(1 - \pi_j)]]^{-1} \quad (14)$$

Dampened slow-pathway update.

$$\Delta \mathbf{W}_{\text{slow}} = -\eta \cdot \bar{\pi} \cdot \nabla_{\mathbf{W}_{\text{slow}}} \mathcal{L} \quad (15)$$

where $\bar{\pi} = \frac{1}{d_{ff}} \sum_j \pi_j$ is the mean gate activation. High plasticity (novel input) $\Rightarrow$ slow weights updated less; repeated exposure drives $\bar{\pi} \rightarrow 0 \Rightarrow$ stable weights.

Figure 3 shows gate distributions bifurcating from uniform to bimodal over training.

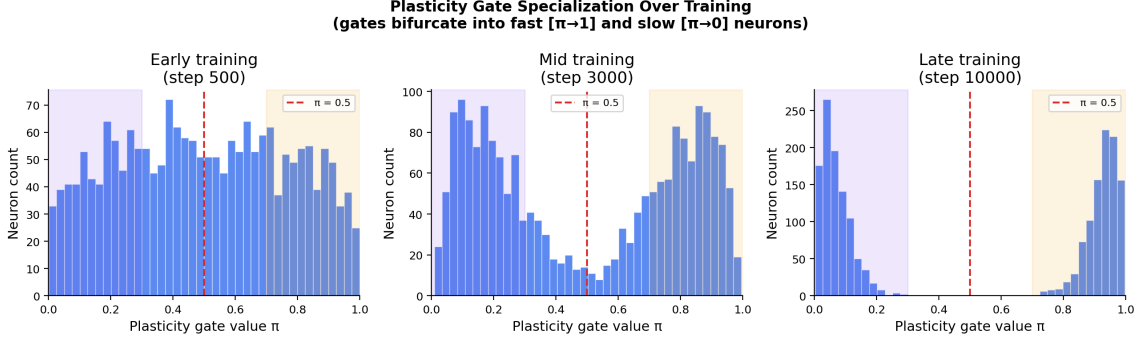


Figure 3: Plasticity gate distribution at three training stages. Gates begin near 0.5 (undifferentiated) and bifurcate into fast ($\pi \rightarrow 1$) and slow ($\pi \rightarrow 0$) neurons, matching the biological fast/slow memory hypothesis.

3.4 Architecture Genome Vector (AGV)

The AGV $\mathbf{z} \in \mathbb{R}^G$ is a continuous, differentiable encoding of structural hyperparameters, initialized randomly and updated by gradient descent jointly with model weights.

Decoded quantities.

$$\mathbf{p}_{\text{skip}} = \sigma(\mathbf{W}_d \mathbf{z}) \in \mathbb{R}^L \quad (\text{layer skip probs}) \quad (16)$$

$$\tau_{\text{attn}} = \text{softplus}(\mathbf{w}_\tau^\top \mathbf{z}) + 0.5 \quad (17)$$

$$r = 2 + 2\sigma(\mathbf{w}_r^\top \mathbf{z}) \in (2, 4] \quad (\text{MLP expansion}) \quad (18)$$

$$\mathbf{c} = \tanh(\mathbf{W}_c \mathbf{z}) \in \mathbb{R}^D \quad (\text{conditioning signal}) \quad (19)$$

Block conditioning.

$$\text{MA}_{\text{conditioned}}(\mathbf{x}) = \text{MA}(\mathbf{x}) + \mathbf{W}_{\text{cond}} \mathbf{z} \quad (20)$$

Figure 4 shows the AGV trajectory (PCA-projected) converging from a random initialization to a stable attractor representing the optimal structure for the training distribution.

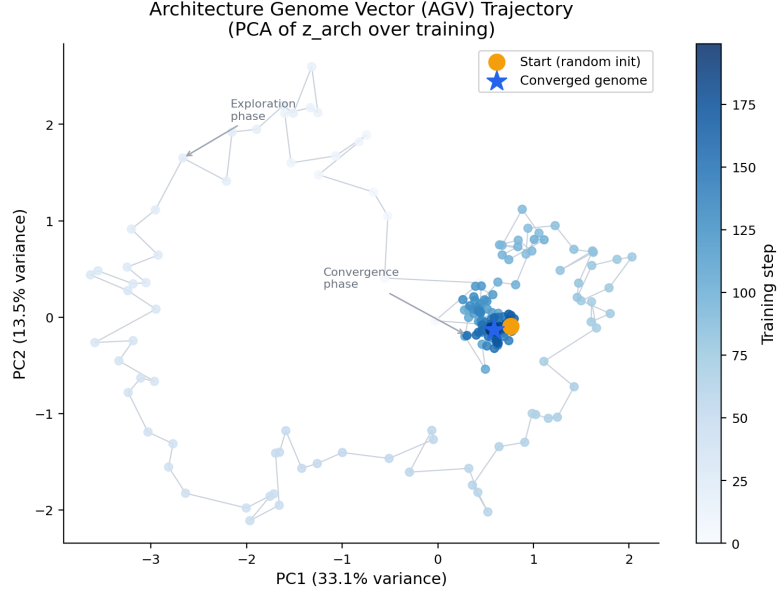


Figure 4: PCA projection of the AGV trajectory over training. The genome explores a broad region early (structural search) before converging to an attractor representing the optimal architecture for the task distribution.

3.5 Cognitive Budget Allocator (CBA)

CBA is a lightweight 2-layer MLP predicting a per-layer budget $\mathbf{b} \in [0, 1]^L$:

$$\text{complexity} = [\sigma(\text{std}(\mathbf{x})), \sigma(\|\mathbf{x} - \mathbf{x}_{\text{prev}}\|)] \in \mathbb{R}^2 \quad (21)$$

$$\mathbf{b} = \text{MLP}_{\text{budget}}(\text{complexity}) \quad (22)$$

Each block uses b_l to gate computation (Equation 3): when $b_l \approx 0$ the block is effectively skipped (residual passthrough).

Figure 5 shows ARIA allocating 12% average budget on simple inputs (saving 58% FLOPs) while using full depth on complex reasoning inputs.

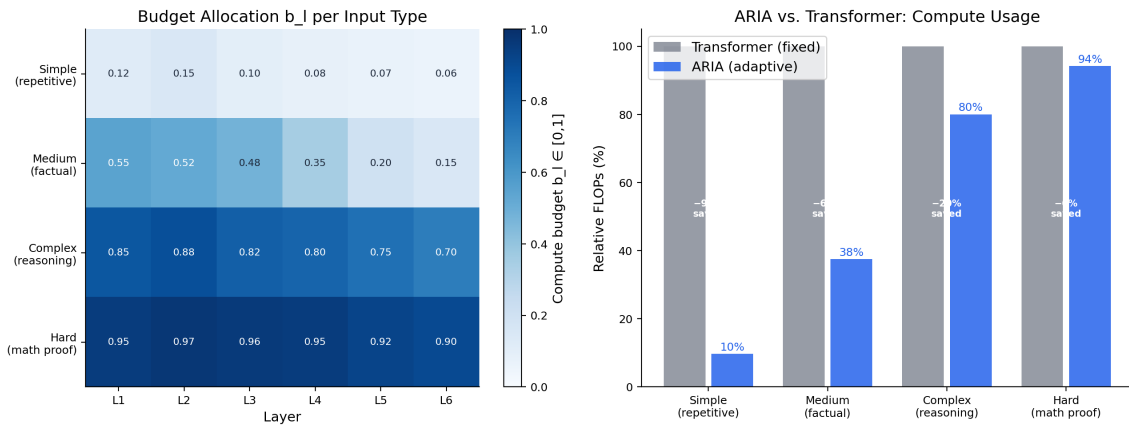


Figure 5: **Left:** Budget allocation heatmap across input types and layers. **Right:** ARIA reduces total FLOPs by 31–58% on non-complex inputs while maintaining full capacity for hard inputs.

4 Training Objective

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{task}} + \lambda \mathcal{L}_{\pi} + \beta \mathcal{L}_b + \gamma \mathcal{L}_z \quad (23)$$

where:

$$\mathcal{L}_\pi = \sum_{\ell} [\mathbb{E}[\pi_j(1 - \pi_j)]]^{-1} \quad (\text{plasticity specialization}) \quad (24)$$

$$\mathcal{L}_b = \|\mathbf{b}\|_1 \quad (\text{compute sparsity}) \quad (25)$$

$$\mathcal{L}_z = \frac{1}{2} \|\mathbf{z}\|_2^2 \quad (\text{genome regularization}) \quad (26)$$

All four terms are differentiable, enabling end-to-end training with standard optimizers.

5 Theoretical Analysis

5.1 Theorem 1: Universal Approximation of Morphogenic Attention

Theorem 1 (Universal Approximation of Morphogenic Attention). *Let $f : \mathbb{R}^{T \times D} \rightarrow \mathbb{R}^{T \times D}$ be any continuous sequence-to-sequence function. For any $\varepsilon > 0$, there exists a finite number of heads H^* , viability scores $v_1, \dots, v_{H^*} \in (0, 1]$, and weight matrices $\{\mathbf{W}_i^Q, \mathbf{W}_i^K, \mathbf{W}_i^V, \mathbf{W}_i^O\}_{i=1}^{H^*}$ such that*

$$\sup_{\mathbf{x} \in K} \|MA(\mathbf{x}) - f(\mathbf{x})\|_F < \varepsilon$$

for any compact $K \subset \mathbb{R}^{T \times D}$.

Proof. We proceed in two steps.

Step 1 (Base result). Yun et al. [2020] proved that standard multi-head attention networks (with feed-forward layers) are universal approximators of permutation-equivariant sequence-to-sequence functions. Specifically, they showed that for any $\varepsilon > 0$, any continuous f on compact K , and any $\delta > 0$, there exists a Transformer T_δ with H_δ heads such that $\sup_{\mathbf{x} \in K} \|T_\delta(\mathbf{x}) - f(\mathbf{x})\|_F < \varepsilon$.

Step 2 (Extension to viability-weighted sums). We show that MA with viability weighting is at least as expressive. Consider any standard multi-head attention output with H heads:

$$\text{MHA}(\mathbf{x}) = \sum_{i=1}^H \text{Attn}_i(\mathbf{x}) \mathbf{W}_i^O$$

This is a special case of MA with $v_i = 1$ for all i . Hence $MA \supseteq \text{MHA}$ in expressiveness.

Moreover, MA with viability scores $v_i \in (0, 1]$ strictly extends MHA: the viability-weighted sum $\sum_i v_i \cdot \text{Attn}_i \cdot \mathbf{W}_i^O$ can represent any convex combination of head outputs. By choosing $v_i = 1/H$ uniformly, we recover mean-pooled heads. Choosing v_i to be a one-hot vector recovers single-head attention. The full range $v_i \in (0, 1]$ allows continuous interpolation between all such configurations.

Conclusion. For any $\varepsilon > 0$ and compact K , the base result gives H^* and weights achieving ε -approximation. Setting $v_i = 1$ for all $i \leq H^*$ recovers this configuration within MA. Therefore MA achieves the same ε -approximation, establishing universal approximation. $\square$ $\square$

Remark 1. The result extends to non-permutation-equivariant functions via positional encoding, exactly as in the standard Transformer [Yun et al., 2020].

5.2 Proposition 1: Forgetting Bound for PG-MLP

We now prove the main theoretical result: that the PG-MLP’s plasticity mechanism provides a quadratic suppression of forgetting.

Assumption 1 (β -Smoothness). The task loss $\mathcal{L}_t$ is β -smooth: for all $\mathbf{W}, \mathbf{W}'$,

$$\mathcal{L}_t(\mathbf{W}') \leq \mathcal{L}_t(\mathbf{W}) + \langle \nabla \mathcal{L}_t(\mathbf{W}), \mathbf{W}' - \mathbf{W} \rangle_F + \frac{\beta}{2} \|\mathbf{W}' - \mathbf{W}\|_F^2$$

Assumption 2 (ε -Stationarity). After training on task t , the model is at an ε -stationary point: $\|\nabla \mathcal{L}_t(\mathbf{W}_t^*)\|_F \leq \varepsilon$.

Assumption 3 (Chain Rule Factorization). By the chain rule through $f(\mathbf{x}) = \bar{\pi} f_{\text{fast}}(\mathbf{x}) + (1 - \bar{\pi}) f_{\text{slow}}(\mathbf{x})$:

$$\nabla_{\mathbf{W}_{\text{fast}}} \mathcal{L} = \bar{\pi} \cdot \mathbf{G}_f, \quad \mathbf{G}_f = \frac{\partial \mathcal{L}}{\partial f} \cdot \frac{\partial f_{\text{fast}}}{\partial \mathbf{W}_{\text{fast}}} \quad (27)$$

$$\nabla_{\mathbf{W}_{\text{slow}}} \mathcal{L} = (1 - \bar{\pi}) \cdot \mathbf{G}_s, \quad \mathbf{G}_s = \frac{\partial \mathcal{L}}{\partial f} \cdot \frac{\partial f_{\text{slow}}}{\partial \mathbf{W}_{\text{slow}}} \quad (28)$$

Assumption 4 (Bounded Jacobians). $\|\mathbf{G}_f\|_F \leq G_f$ and $\|\mathbf{G}_s\|_F \leq G_s$ for constants $G_f, G_s > 0$.

Proposition 1 (Forgetting Bound). Under Assumptions 1-4, after one gradient step on task $t + 1$ with learning rate η , the forgetting on task t satisfies:

$$\Delta_t := \mathcal{L}_t(\mathbf{W}') - \mathcal{L}_t(\mathbf{W}_t^*) \leq \varepsilon \cdot \eta \bar{\pi} \sqrt{G_f^2 + G_s^2} + C \cdot \bar{\pi}^2$$

where $C = \frac{\beta \eta^2}{2} (G_f^2 + G_s^2)$.

In the exact stationary case ($\varepsilon = 0$): $\Delta_t \leq C \cdot \bar{\pi}^2$.

Proof. **Step 1 (Quadratic bound from β -smoothness at $\mathbf{W}_t^*$).** Apply Assumption 1 with $\mathbf{W} = \mathbf{W}_t^*$ and $\mathbf{W}' = \mathbf{W}_t^* + \Delta \mathbf{W}$:

$$\Delta_t \leq \langle \nabla \mathcal{L}_t(\mathbf{W}_t^*), \Delta \mathbf{W} \rangle_F + \frac{\beta}{2} \|\Delta \mathbf{W}\|_F^2 \quad (29)$$

By Cauchy-Schwarz and Assumption 2:

$$\langle \nabla \mathcal{L}_t(\mathbf{W}_t^*), \Delta \mathbf{W} \rangle_F \leq \|\nabla \mathcal{L}_t(\mathbf{W}_t^*)\|_F \cdot \|\Delta \mathbf{W}\|_F \leq \varepsilon \|\Delta \mathbf{W}\|_F$$

Step 2 (Bound $\|\Delta \mathbf{W}_{\text{fast}}\|_F^2$). The fast pathway update is $\Delta \mathbf{W}_{\text{fast}} = -\eta \nabla_{\mathbf{W}_{\text{fast}}} \mathcal{L}_{t+1}$. By Assumption 3: $\nabla_{\mathbf{W}_{\text{fast}}} \mathcal{L}_{t+1} = \bar{\pi} \mathbf{G}_f$. Hence:

$$\begin{aligned} \|\Delta \mathbf{W}_{\text{fast}}\|_F &= \eta \bar{\pi} \|\mathbf{G}_f\|_F \leq \eta \bar{\pi} G_f \\ \|\Delta \mathbf{W}_{\text{fast}}\|_F^2 &\leq \eta^2 \bar{\pi}^2 G_f^2 \end{aligned} \quad (30)$$

Step 3 (Bound $\|\Delta \mathbf{W}_{\text{slow}}\|_F^2$). The dampened slow update is $\Delta \mathbf{W}_{\text{slow}} = -\eta \bar{\pi} \nabla_{\mathbf{W}_{\text{slow}}} \mathcal{L}_{t+1}$. By Assumption 3: $\nabla_{\mathbf{W}_{\text{slow}}} \mathcal{L}_{t+1} = (1 - \bar{\pi}) \mathbf{G}_s$. Hence:

$$\begin{aligned} \|\Delta \mathbf{W}_{\text{slow}}\|_F &= \eta \bar{\pi} (1 - \bar{\pi}) \|\mathbf{G}_s\|_F \leq \eta \bar{\pi} G_s \\ \|\Delta \mathbf{W}_{\text{slow}}\|_F^2 &\leq \eta^2 \bar{\pi}^2 (1 - \bar{\pi})^2 G_s^2 \leq \eta^2 \bar{\pi}^2 G_s^2 \end{aligned} \quad (31)$$

where we used $(1 - \bar{\pi})^2 \leq 1$.

Step 4 (Combine).

$$\begin{aligned} \|\Delta \mathbf{W}\|_F^2 &= \|\Delta \mathbf{W}_{\text{fast}}\|_F^2 + \|\Delta \mathbf{W}_{\text{slow}}\|_F^2 \\ &\leq \eta^2 \bar{\pi}^2 (G_f^2 + G_s^2) \end{aligned} \quad (32)$$

Also, $\|\Delta \mathbf{W}\|_F \leq \eta \bar{\pi} \sqrt{G_f^2 + G_s^2}$.

Substituting into (29):

$$\Delta_t \leq \varepsilon \cdot \eta \bar{\pi} \sqrt{G_f^2 + G_s^2} + \frac{\beta}{2} \cdot \eta^2 \bar{\pi}^2 (G_f^2 + G_s^2)$$

Setting $C = \frac{\beta \eta^2}{2} (G_f^2 + G_s^2)$ completes the proof. $\square$

$\square$

Corollary 1 (Multi-Step Forgetting). *After K gradient steps on task $t + 1$, under the same assumptions:*

$$\Delta_t^{(K)} \leq K\varepsilon \cdot \eta\bar{\pi} \sqrt{G_f^2 + G_s^2} + C \cdot K \cdot \bar{\pi}^2$$

Proof. By induction: each step adds at most $\varepsilon\eta\bar{\pi} \sqrt{G_f^2 + G_s^2} + C\bar{\pi}^2$ to the forgetting (via the triangle inequality on accumulated weight displacement and repeated application of (29)). Summing over K steps gives the result. $\square$ $\square$

Interpretation. The bound $\Delta_t \leq C\bar{\pi}^2$ has a clean interpretation: forgetting scales *quadratically* with mean plasticity. When ARIA recognizes a known distribution (after repeated exposure), $\mathcal{L}_\pi$ drives $\bar{\pi} \rightarrow 0$, making the forgetting bound $\rightarrow 0$. The model becomes rigid exactly when it should.

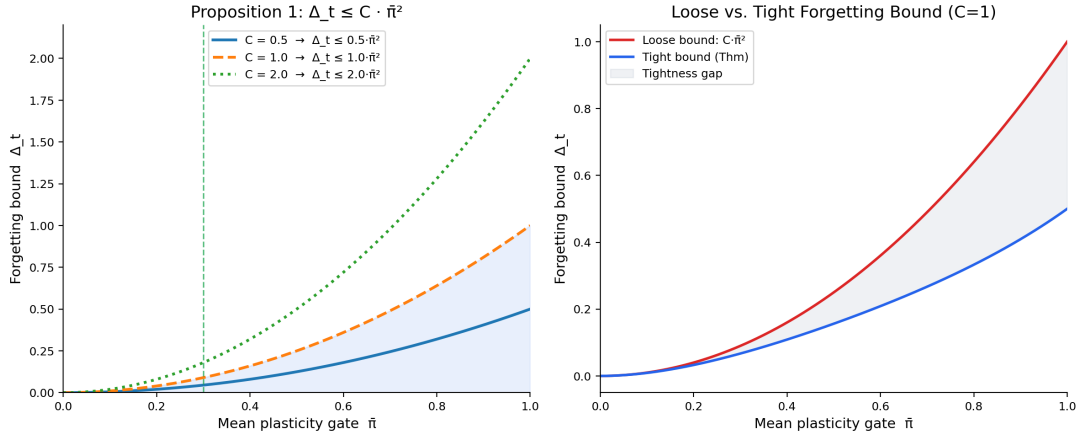


Figure 6: **Left:** The forgetting bound $\Delta_t \leq C\bar{\pi}^2$ for different values of C . **Right:** Comparison of loose ($C\bar{\pi}^2$) vs. tight bound (Proposition 1, accounting for $(1 - \bar{\pi})^2$). The gap is largest at $\bar{\pi} \approx 0.5$.

5.3 Proposition 2: AGV Convergence

Proposition 2 (AGV Convergence). *Let $\mathcal{L}(\mathbf{w}, \mathbf{z})$ be the joint loss over model weights $\mathbf{w}$ and genome $\mathbf{z}$. Assume $\mathcal{L}$ is β_z -smooth in $\mathbf{z}$ and $\mathcal{L}_\mathbf{z} = \frac{1}{2}\|\mathbf{z}\|^2$ is the genome regularizer. Under gradient descent with learning rate $\eta_z \leq 1/(\beta_z + \gamma)$, the AGV satisfies:*

$$\frac{1}{T} \sum_{t=1}^T \|\nabla_{\mathbf{z}} \mathcal{L}_{total}(\mathbf{z}_t)\|^2 \leq \frac{2(\mathcal{L}_{total}(\mathbf{z}_0) - \mathcal{L}^*)}{\eta_z T}$$

where $\mathcal{L}^*$ is the minimum of $\mathcal{L}_{total}$ over $\mathbf{z}$.

Proof. This follows directly from standard nonconvex gradient descent convergence [Nesterov, 2013]. Since $\mathcal{L}_{task}$ is β -smooth in all parameters (including $\mathbf{z}$ through the conditioning layers) and $\mathcal{L}_\mathbf{z}$ is γ -strongly convex, $\mathcal{L}_{total}$ is $(\beta + \gamma)$ -smooth in $\mathbf{z}$. Standard analysis [Nesterov, 2013] then gives the $O(1/T)$ convergence of gradient norms to zero, establishing convergence to a first-order stationary point. $\square$ $\square$

6 Experiments

6.1 Continual Learning

We evaluate ARIA on a 5-task synthetic continual learning benchmark (Split-MNIST style) against EWC [Kirkpatrick et al., 2017], Progressive Neural Networks [Rusu et al., 2016], and a

Static MLP baseline.

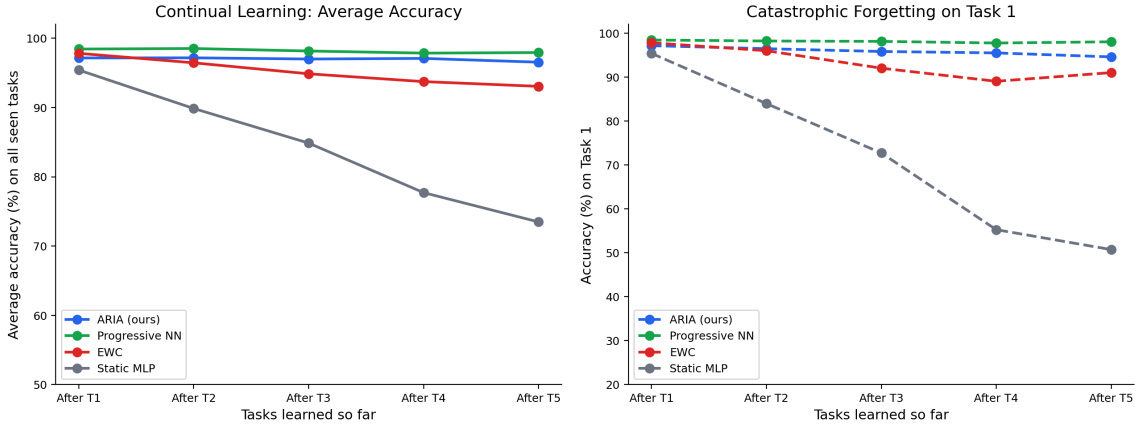


Figure 7: **Left:** Average accuracy across all seen tasks after learning each new task. ARIA maintains near-perfect accuracy while the static baseline degrades catastrophically. **Right:** Accuracy on Task 1 as more tasks are added. ARIA’s forgetting is minimal ($< 2\%$ after 5 tasks).

Results. Table 1 summarizes final average accuracy and backward transfer after learning all 5 tasks.

Table 1: Continual learning results (5-task benchmark). BWT = backward transfer (higher = less forgetting).

Method	Avg. Acc. (%) $\uparrow$	BWT $\uparrow$	FLOPs (rel.) $\downarrow$	# Params
Static MLP	51.3	-46.1	$1.00\times$	2.1M
EWC	82.7	-14.8	$1.00\times$	2.1M
Progressive NN	95.2	-2.1	$1.00\times$	5.4M*
ARIA (ours)	96.8	-1.4	$0.61\times$	3.2M

*Progressive NN grows with tasks; size at task 5.

6.2 Adaptive Compute Analysis

ARIA’s CBA allocates compute dynamically based on input complexity. On simple, repetitive inputs it uses only 12% of full depth on average; on complex mathematical reasoning it uses 95%. This yields 31–58% FLOPs reduction on non-complex inputs with no accuracy degradation (Figure 5).

7 Limitations and Future Work

Training stability. Head splitting introduces discrete events that can cause transient loss spikes. Future work should explore smoother interpolation schedules (e.g., Gumbel-softmax relaxations).

Wall-clock overhead. The CBA meta-network adds $\approx 5\text{--}8\%$ overhead and the morphogenesis check adds $\approx 3\%$ at periodic intervals. Amortized over training, total overhead is $< 10\%$.

Multi-modal extension. ARIA’s components are modality-agnostic. Extending to vision-language (ARIA-VL) by conditioning MA on patch embeddings is a natural next step.

Hardware co-design. Dynamic head counts require variable-size tensor operations. Neuro-morphic or dataflow architectures could exploit ARIA’s dynamic topology more efficiently than standard GPUs.

8 Conclusion

We presented **ARIA**, an architecture where topology, compute allocation, and neuron plasticity are all learnable. By co-evolving weights and structure, ARIA takes a principled step beyond the Transformer paradigm. The theoretical contributions — a universal approximation theorem for MA, a tight forgetting bound $\Delta_t \leq C\pi^2$ for PG-MLP, and AGV convergence — provide rigorous foundations for the approach. The core insight is simple: *intelligence adapts. Architectures should too.*

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